

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

Name: Edona Halilaj

2025-11-11

Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

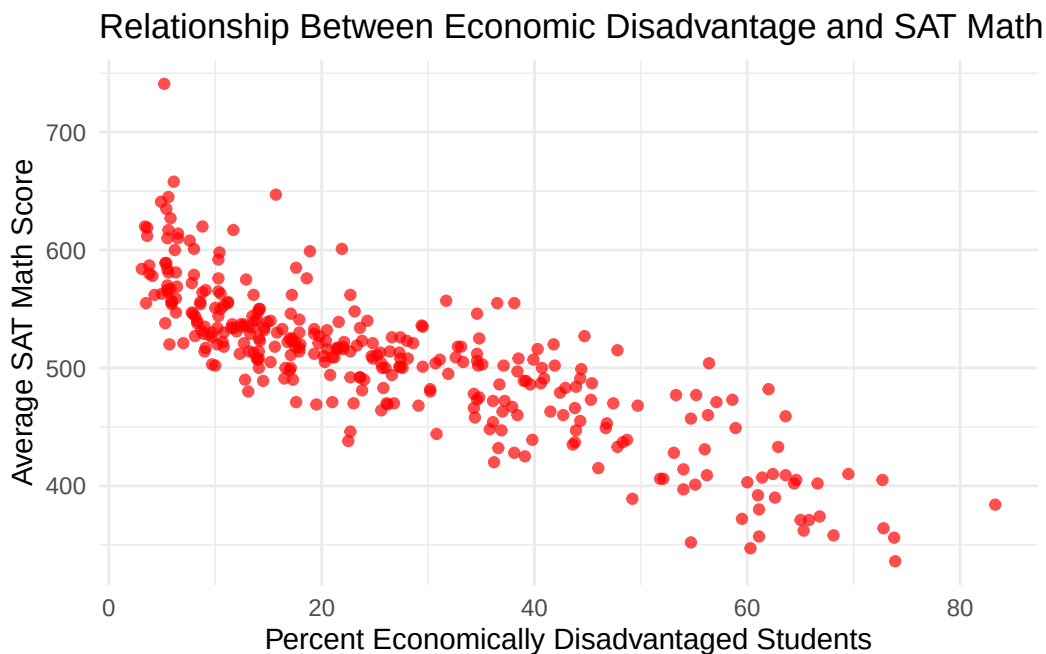
# Load the MA_schools dataset
data(MA_schools)
```

Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
ggplot(MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +  
  geom_point(color = "red", alpha = 0.7) +  
  labs(  
    title = "Relationship Between Economic Disadvantage and SAT Math Scores",  
    x = "Percent Economically Disadvantaged Students",  
    y = "Average SAT Math Score"  
  ) +  
  theme_minimal()
```



Answer: We have a negative relationship between `perc_disadvan` and `average_sat_math`.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
model1 <- lm( average_sat_math ~ perc_disadvan , data=MA_schools)
summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
sat_effects <- coef(model1)
sat_effects
```

(Intercept)	perc_disadvan
581.281054	-2.779787

Intercept: 581.281054 around 581

Slope: -2.779787 around -2.78

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: For every 1 percentage point increase in the proportion of economically disadvantaged students, the average SAT math score is expected to decrease by about 2.78 points, on average, holding all else constant.

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
school <- data.frame(perc_disadvan = 50)
predicted_score <- predict(model1, newdata = school)
predicted_score
```

```
      1
442.2917
```

Predicted average SAT math score: 442

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
predict(model1, newdata = data.frame(perc_disadvan = 50), interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
model2 <- lm( average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-81.342	-20.802	-3.872	18.211	167.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	588.1898	7.6067	77.325	<2e-16 ***
perc_disadvan	-2.7772	0.1063	-26.120	<2e-16 ***
sizemedium	-11.9134	7.5346	-1.581	0.115
sizelarge	-6.3615	6.9231	-0.919	0.359

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
summary(model1)$r.squared
```

```
[1] 0.6961948
```

```
summary(model2)$r.squared
```

```
[1] 0.6987587
```

Answer: Model1 $R^2 = 0.6961948$ About 69.6% of the variance in average SAT math scores is explained by the percentage of economically disadvantaged students. Model2 $R^2 = 0.6987587$ About 69.9% of the variance is explained when adding school size. Adding size on model2 explains only about 0.3% more variance, which is very small change.

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
anova(model1, model2)
```

Analysis of Variance Table

```
Model 1: average_sat_math ~ perc_disadvan
Model 2: average_sat_math ~ perc_disadvan + size
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     330 371191
2     328 368058  2    3132.6 1.3958 0.2491
```

Conclusion: The F-statistic for comparing the two models is 1.3958, with a p-value of 0.2491 which is greater than $\alpha = 0.05$, meaning there's no statistically significant improvement when adding the variable size to the model. Adding school size (size) to the regression model does not significantly improve the model's ability to predict average SAT math scores more than what is explained by the percentage of economically disadvantaged students.